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1 Introduction

1.1 System Introduction

The **Event Management Website** is a web-based platform that allows users to easily explore, register for, and manage various events such as seminars, concerts, workshops, and conferences. It simplifies event management for both organizers and participants by handling registrations, scheduling, and automated notifications through **email, SMS, and WhatsApp** after successful registration. The system's main purpose is to make event handling smooth, user-friendly, and fully digital.

1.2 Background of the System

Existing platforms like Bookkaru mainly focus on event ticketing and sales for commercial purposes, often with high service fees and limited customization. Our Event Management Website is built for universities, colleges, and small organizations that want to create and manage events easily without paying commissions. It is different because it provides simple event creation, participant management, and multi-channel notifications (email, SMS, WhatsApp) in one system. It's designed to be lightweight, free, and adaptable to different event types.

1.3 Objectives of the System

- To provide an online platform for creating and managing events.
- To allow users to register for events and get confirmation instantly.
- To send automatic notifications via **email, SMS, and WhatsApp**.
- To help admins organize event details, participants, and schedules.
- To maintain a secure database of users and event records.
- To make event registration fast, transparent, and easy to track.

1.4 Significance of the System

This system removes the need for manual event registration and paperwork. It can be used for:

- **University and College Events** (seminars, workshops, orientations)
- **Concerts and Social Gatherings**
- **Corporate Training Programs**
- **Community Events and Conferences**

It's important because it saves time, prevents errors, and keeps all event-related data organized and accessible from anywhere.

2 Overall Description

2.1 Product Perspective

The Event Management Website is a new, standalone system designed to manage all event-related operations. It connects users (participants) and organizers through a single platform. It also interacts with SMTP servers (for emails) and SMS/WhatsApp APIs for sending confirmations and updates.

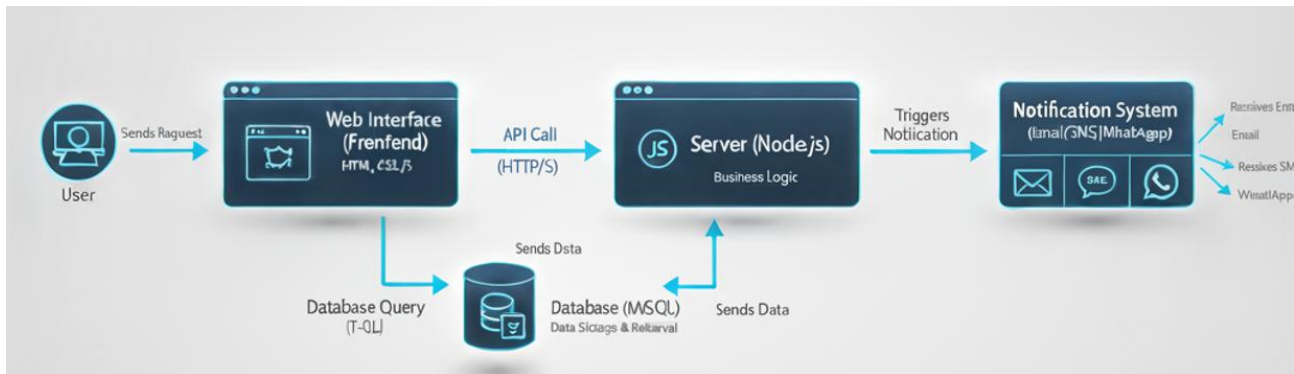


Fig-2.1 General Flow Diagram of System

2.2 Product Scope

The Event Management Website is an online platform that helps organizers create and manage events while allowing users to view details and register easily. It automates confirmation messages through email, SMS, and WhatsApp and securely stores all events and participant data.

The system focuses on event registration, management, and notifications. It does not include online payment, ticket pricing, or promotional features in this version. Future upgrades may add these functions to extend their capabilities.

2.3 Product Functionality

- User registration and login.
- Admin panel for creating and managing events.
- Event search and filtering.

- Participant registration form.
- Automated email/SMS/WhatsApp notifications.
- Database for storing events and participant data.
- Event update and deletion options for admins.

2.4 Users and Characteristics

1. Admin:

The Admin has the highest level of control in the system. They can approve or reject events created by organizers, manage all users, and monitor overall system activity. Admins ensure that only verified and valid events are published on the platform.

2. Organizer:

Organizers can create, edit, and manage their own events after admin approval. They can view participant lists, update event details, and send updates or reminders to register users.

3. User (Participant):

Users can browse approved events, view event details, and register online. After registration, they receive confirmation through email, SMS, or WhatsApp.

All interfaces are designed to be simple, responsive, and user-friendly, ensuring that even users with basic technical knowledge can easily interact with the system.

2.5 Operating Environment

- **Frontend:** HTML, CSS, JavaScript
- **Backend:** Node.js (Express Framework)
- **Database:** MSSQL Server
- **Operating System:** Windows / Linux
- **Other Requirements:** Internet connection, modern web browser (Chrome, Edge, Firefox)

3 Specific Requirements

3.1 Functional Requirements

3.1.1 Admin Functions

1. User Management:

- View, edit, and delete user accounts.
- Approve or block organizers.

2. Event Approval:

- Review events submitted by organizers.
- Approve or reject events before publishing.

3. System Monitoring:

- View total users, events, and registrations in the dashboard.
- Generate reports for overall activity.

4. Security Control:

- Manage login credentials and enforce secure authentication.

3.1.2 Organizer Functions

1. Event Creation and Management:

- Create new events with details like title, date, venue, and description.
- Edit or delete events before or after approval.

2. Participant Management:

- View lists of registered participants.
- Send event updates or reminders through notifications.

3. Profile Management:

- Update organizer profile and contact details.

3.1.3 User (Participant) Functions

1. Registration and Login:

- Create an account and securely log in to the system.

2. Event Browsing:

- View all approved events with filters for date, type, or location.

3. Event Registration:

- Register for any active event.
- Receive confirmation through email, SMS, or WhatsApp.

4. Profile Management:

- Update personal details and view past registrations.

3.2 Behaviour Requirements

3.2.1 Admin – Use Cases

1. **Approve / Reject Events** – Review events submitted by organizers and decide their status.
2. **Manage Users** – Add, edit, block, or delete users and organizers.
3. **View Reports / Dashboard** – See system statistics like total events, users, and registrations.
4. **Send Notifications** – Communicate important updates or alerts to organizers or users.

3.2.2 Organizer – Use Cases

1. **Create Event** – Add new events with details (title, date, venue, description, etc.).
2. **Edit / Delete Event** – Modify or remove events they have created.
3. **View Participants** – See the list of users registered for their events.
4. **Send Notifications** – Send updates or reminders to participants.

3.2.3 User (Participant) – Use Cases

1. **View Events** – Browse all approved events by date, category, or location.
2. **Register for Event** – Fill out the registration form to join an event.
3. **Receive Notifications** – Get confirmation or update messages via email, SMS, or WhatsApp.

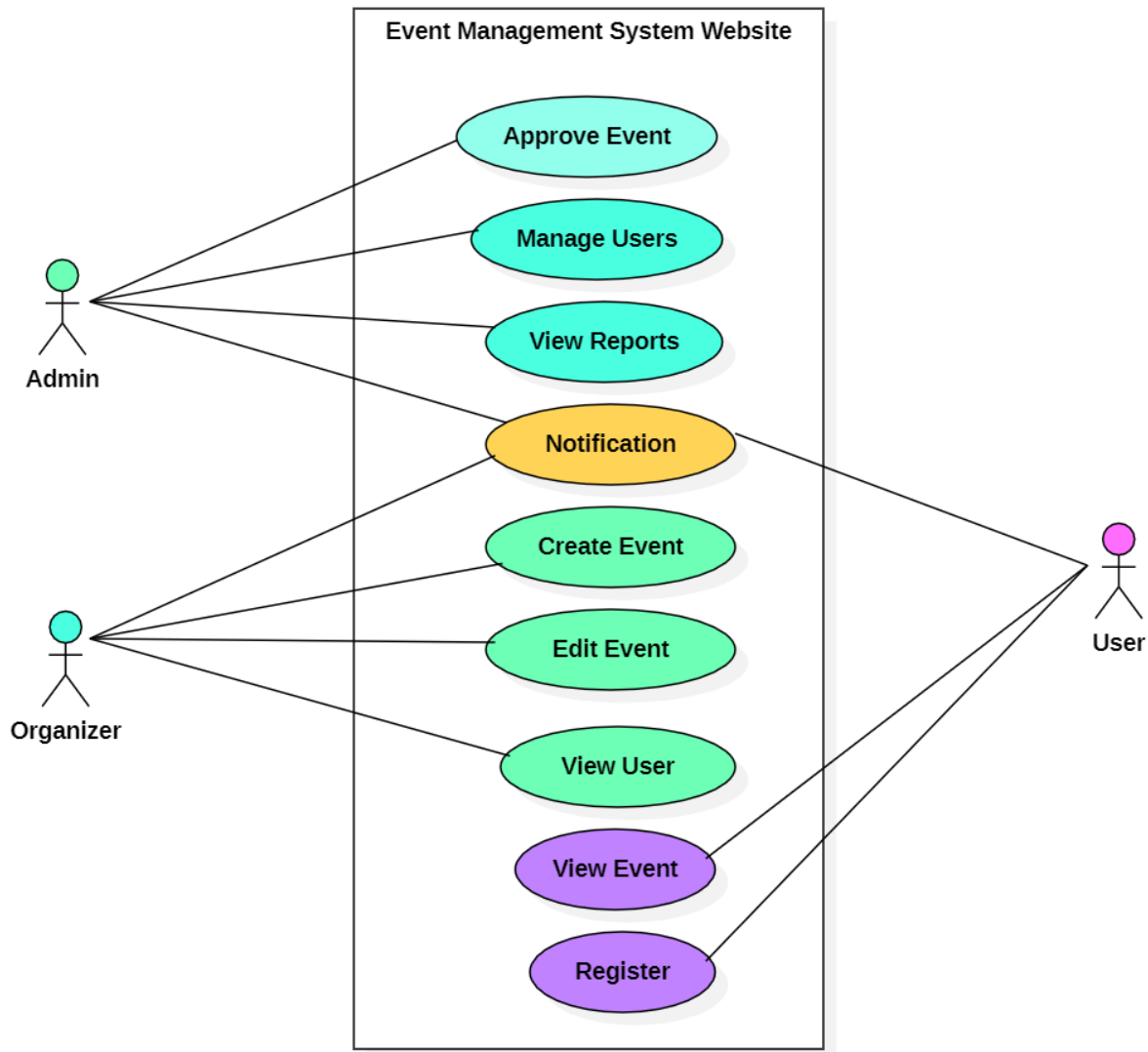


Fig-3.1 Use Case Diagram

3.3 External Interface Requirements

3.3.1 User Interfaces

The **Event Management Website** provides a clean, responsive, and user-friendly interface for all three types of users — **Admin**, **Organizer**, and **User (Participant)**.

3.3.1.1 Main Interfaces:

- **Login / Registration Page:**

Allows users, organizers, and admins to log in securely or sign up. It includes fields for email, password, and role selection. Error messages appear for invalid credentials.

- **Home Page:**

Displays upcoming approved events with details like event name, date, venue, and description. Users can filter or search events easily.

- **Event Details Page:**

Shows complete event information with a “Register Now” button for users.

- **Registration Form:**

Collects user details for event registration and displays confirmation after submission.

- **Admin Dashboard:**

Includes event approval panel, user management section, and system overview cards showing total events, organizers, and participants.

- **Organizer Dashboard:**

Contains options to create, edit, and delete events, view participants, and send notifications.

3.3.1.2 Standard Elements:

- **Buttons:** Submit, Cancel, Update, Delete, Approve, Reject
- **Alerts:** Success or error messages displayed in pop-up or inline formats
- **Navigation:** Sidebar and top bar for easy access across sections

3.3.2 Hardware Interfaces

The system does not require any special hardware. It runs on standard computing devices such as desktops, laptops, or smartphones with an active internet connection. A basic keyboard and mouse (or touchscreen) are sufficient for all interactions.

No specialized hardware libraries or interfaces are needed for this project.

3.3.3 Software Interfaces

The system interacts with the following software components:

- **Operating System:** Compatible with Windows, Linux, and macOS.

- **Database:** MSSQL Server (for storing users, events, and registrations).
- **Backend:** Node.js (Express framework) for handling requests and logic.
- **Frontend:** HTML, CSS, and JavaScript for the user interface.

3.3.4 Communications Interfaces

The system communicates using standard HTTP/HTTPS protocols between client and server. All user requests and responses are sent securely using HTTPS to protect data in transit.

For notifications, the system uses:

- **SMTP protocol** for sending emails via a free email service (e.g., Gmail SMTP or Sendinblue free plan).
- **Firebase Cloud Messaging (FCM)** or **OneSignal** for push notifications instead of paid SMS/WhatsApp services.

The system uses **Microsoft SQL Server (MSSQL)** for database management, ensuring secure and reliable data storage.

All data transfers are encrypted to ensure privacy and security. The system requires an active internet connection for proper synchronization between the web server, database, and notification services.